UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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PART A
QUESTION 1

Refer to the question items below.

a) [THEORY SLOT - attacks/malware, 2 marks, short objective]

(2 marks)

b) [THEORY SLOT - attacks/malware, 6 marks, define/explain]

(6 marks)

QUESTION 2

Refer to the question items below.

a) Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
KLCPJLYLABYULKHUKDPWLKVUL

(2 marks)

b) Encrypt the message below using Playfair encryption with the key REDFOX. Use a 5 x 5 grid, merge I and J, and use X as the filler between repeated letters and at the end if needed.
REVIEW ACCESS PERMISSIONS

(6 marks)

QUESTION 3

Assume a computer system with the accounts: Ali, Yati, Umi, Root. Ali can read, write X-1; Yati can read X-1; Umi can execute REPORT; Root can read, write, execute DATA.

a) [THEORY SLOT - access control models, 2 marks, short objective]

(2 marks)

b) Based on the scenario above, write the access control matrix (ACM), the access control lists (ACL) and the capability lists (CL), and draw the diagram.

(6 marks)

QUESTION 4

Refer to the question items below.

a) Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
AYHPULTWSVFLLZVUZLJBYPAF

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(2 marks)

a) Encrypt the message below using Playfair encryption with the key REDFOX. Use a 5 x 5 grid, merge I and J, and use X as the filler between repeated letters and at the end if needed.
CONTAINER SECURITY CHECKS

(6 marks)

QUESTION 5

Refer to the question items below.

a) [THEORY SLOT - authentication/biometrics, 2 marks, short objective]

(2 marks)

b) Decrypt the following ciphertext using columnar transposition with a 3 x 4 matrix (3 rows, 4 columns). Put the ciphertext into the columns, read the plaintext out of the rows, and ignore the filler X at the end. The key is the column size, 4.
LNIOATGUXIDX

(6 marks)

QUESTION 6

Refer to the question items below.

a) [THEORY SLOT - software/OS security, 2 marks, short objective]

(2 marks)

b) [THEORY SLOT - software/OS security, 6 marks, define/explain]

(6 marks)

QUESTION 7

Refer to the question items below.

a) [THEORY SLOT - network security, 2 marks, short objective]

(2 marks)

b) [THEORY SLOT - network security, 6 marks, define/explain]

(6 marks)

QUESTION 8

Refer to the question items below.

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a) [THEORY SLOT - trusted computing, 2 marks, short objective]

(2 marks)

b) [THEORY SLOT - trusted computing, 6 marks, define/explain]

(6 marks)

QUESTION 9

Refer to the question items below.

a) [THEORY SLOT - authentication/biometrics, 2 marks, short objective]

(2 marks)

b) [THEORY SLOT - authentication/biometrics, 6 marks, define/explain]

(6 marks)

QUESTION 10

Refer to the question items below.

a) [THEORY SLOT - attacks/malware, 2 marks, short objective]

(2 marks)

b) [THEORY SLOT - attacks/malware, 6 marks, define/explain]

(6 marks)

PART B
QUESTION 1

Refer to the scenario.

a) [THEORY SLOT - attacks/malware, 5 marks, scenario/justify]

(5 marks)

b) [THEORY SLOT - attacks/malware, 5 marks, scenario/justify]

(5 marks)

QUESTION 2

Refer to the scenario.

a) [THEORY SLOT - authentication/biometrics, 5 marks, scenario/justify]

(5 marks)

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b) [THEORY SLOT - authentication/biometrics, 5 marks, scenario/justify]

(5 marks)

END OF QUESTION PAPER

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